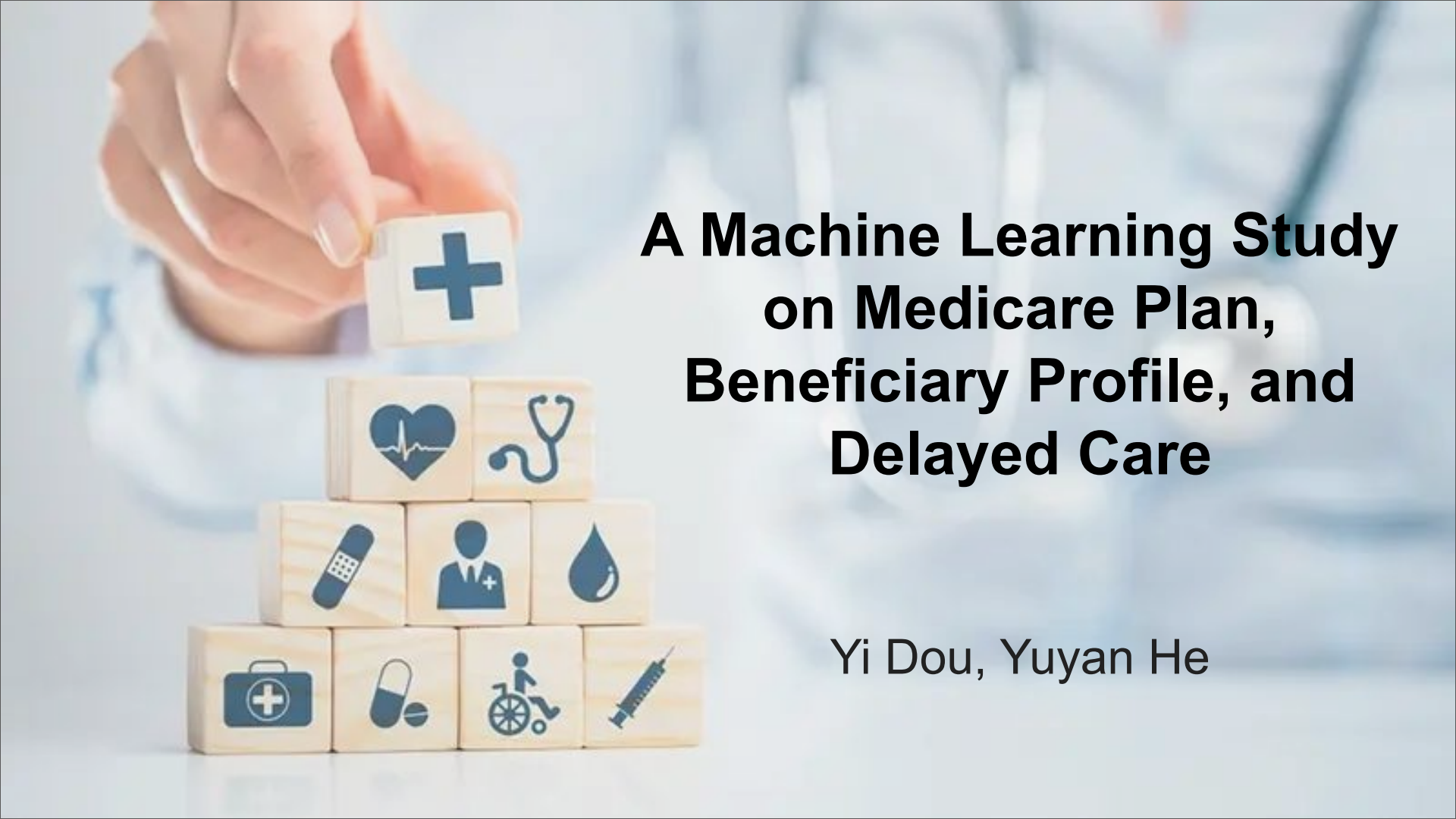


A Machine Learning Study on Medicare Plan, Beneficiary Profile, and Delayed Care

Yi Dou, Yuyan He

Background

Study Objective: Do Medicare enrollment types (Medicare Advantage & Traditional Medicare) concentrate specific beneficiary groups, and does the types affect healthcare behavior, specifically, delayed care due to cost?

Data

Medicare Current Beneficiary Survey
Fall 2023 — 13,016 respondents, 244 variables

Methodology

Unsupervised Learning

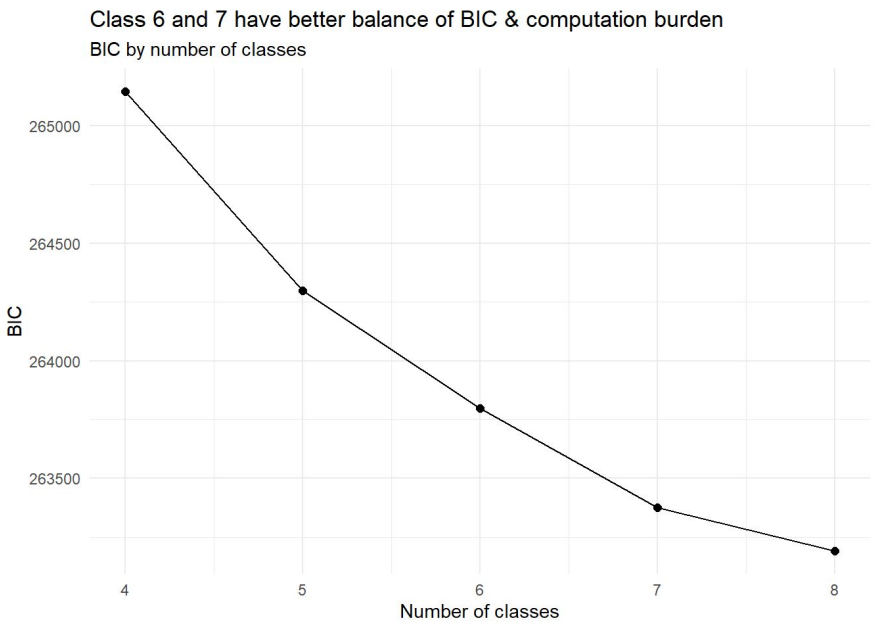
Latent Class Analysis clusters beneficiaries by demographic, socioeconomic, and health characteristics

Supervised Learning

Regression tree, Random Forest, XGBoost, and stacked ensemble, predicting delayed care

Latent Class Analysis Results

Selected a small set of variables that describe baseline demographic, socioeconomic, housing, family, and health-related characteristics



Class 1: Strained Traditionalists
Mixed enrollment (MA 48%), not poor, non-dual, moderate health, high-need, high delayed care

Class 2: Secure Mainstream Beneficiaries
Slightly TM-leaning (MA 42%), healthiest, not poor, non-dual, low-need, lowest delayed care

Class 3: High-Risk Young Disadvantaged
MA-dominant (69%), severely vulnerable, youngest, poor, very unhealthy, dual-eligible, highest-need, high delayed care

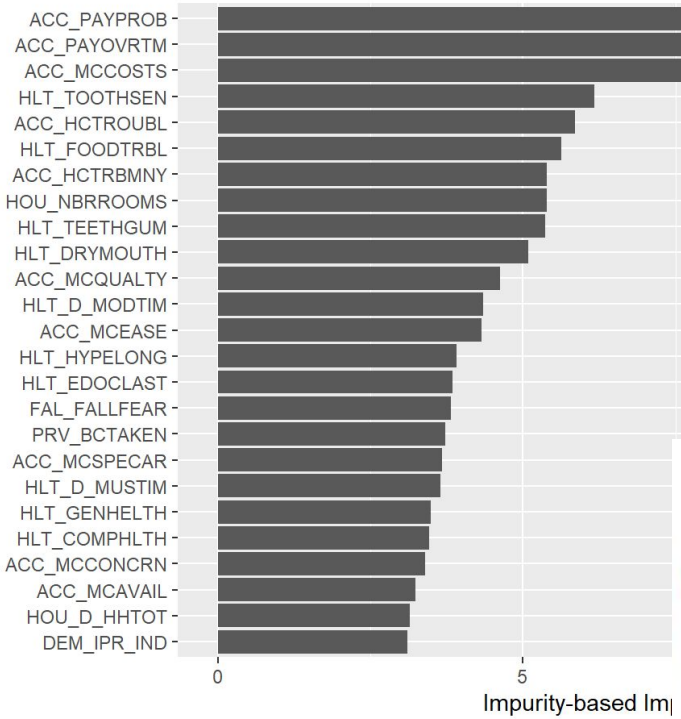
Class 4: Fragile Senior
MA-dominant (77%), older, poor, dual-eligible, moderate-need & health, low delayed care

Class 5: Younger Healthy Poor
MA-dominant (67%), younger, poor, dual-eligible, with moderate need, low delayed care

Class 6: Stable Healthy Seniors
MA-dominant (69%), not poor, non-dual eligible, older but relatively healthy, low delayed care

Predictive Model: Predicting Delay

Variable Screening: Top 25 Predictors



model	auc	recall	precision	f1
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1 Decision Tree	0.715	0.635	0.188	0.289
2 Random Forest	0.834	0.237	0.649	0.347
3 XGBoost	0.851	0.679	0.249	0.365
4 Stacked Ensemble	0.849	0.167	0.703	0.269

model	threshold	auc	recall	precision	f1
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1 Decision Tree	0.84	0.715	0.359	0.389	0.373
2 Random Forest	0.37	0.834	0.487	0.388	0.432
3 XGBoost	0.73	0.851	0.526	0.414	0.463
4 Stacked Ensemble	0.2	0.849	0.551	0.398	0.462